

Shreenivas Rangarajan

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EDUCATION

McMaster University MASc in Mechanical Engineering Coursework: Advanced Fluid Mechanics, Numerical Analysis, Turbulent Flows, Machine Learning	Sep 2023 - Apr 2026 GPA: 3.56/4
Shiv Nadar University BTech in Mechanical Engineering (with Distinction) Coursework: Fluid Mechanics, Mechanical Vibrations, Finite Element Methods, Computational Techniques	Aug 2018 - May 2022 CGPA: 8.58/10

EXPERIENCE

McMaster University & University of Waterloo Graduate Research PI: Dr. Chris Morton, Dr. Serhiy Yarusevych	Sep 2023 - Apr 2026 Hamilton & Waterloo, ON
- Executed two experimental campaigns investigating fluid-structure interaction (FSI) for cylindrical bluff bodies, utilising high-precision Particle Image Velocimetry (PIV) to characterise complex wake dynamics. - Optimised passive geometric modification on a cylinder to achieve over 5% drag and 50% lift force reduction across various turbulence levels using POD analysis of the wake velocity data. - Uncovered the source for amplitude differences in hysteretic regime transitions for vortex-induced vibration (VIV) systems with energy and spectral analysis of the forcing and wake data.	
Shiv Nadar University UG Research Assistant PI: Dr. J. Venkatramani	Sep 2021 - Mar 2023 India
- Discovered the susceptibility of a pitch-plunge airfoil system to prematurely lose stability and undergo stall flutter via transient growth. - Identified the non-normal nature of the system using linear stability and pseudospectral analysis. - Developed a semi-empirical model based on dynamic stall model to simulate experimentally obtained aeroelasticity data.	
Paanduv CFD Research Intern	Mar 2022 - May 2022 India
Developed a multiphase solver for simulating photocatalytic water splitting reaction for hydrogen generation on OpenFOAM and augmented the reaction rate based on the incident light intensity.	
Shiv Nadar University UG Research Assistant PI: Dr. Sathi Reddy	Oct 2020 - Mar 2021 India
- Performed CFD simulation of cavitating flows around hydrofoil on OpenFOAM using a custom solver for multiphase flows based on the Volume of Fluid method. - Compared results from RAS and LES methods for different cavitation models to obtain the ideal angle of attack for limiting cavitation.	

TEAM PROJECTS

ASME IAM3D Competition Team Captain	Oct 2019 - Feb 2020 India
- Mentored a team of 7 freshmen to build an agile 3D printed drone capable of carrying a payload. - Optimised the design using CFD and FEA simulations for maneuverability and flight time.	

SKILLS

- CAD/CAE software:** SolidWorks, OpenFOAM, ANSYS Fluent, ANSYS Mechanical
- Programming Languages and Packages:** MATLAB, Python, C/C++, Tensorflow, PyTorch, L^AT_EX

PUBLICATIONS

- Shreenivas R.** et al. "Non-normality and transient growth in stall flutter instability". Chaos (AIP) (2023).
- Dheeraj T., **Shreenivas R.** et al. "Experimental investigation on the synchronization characteristics of a pitch-plunge aeroelastic system exhibiting stall flutter". Chaos (AIP) (2022).